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Evidence-Preserving Research Amortization:

A Formal Cost Model and Experimental Agenda for Shared Knowledge in AI-Agent Systems

Signal Atlas Research Program

Independent research proposal • No institutional affiliation claimed

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Theoretical contribution; empirical validation pending

ABSTRACT

Contemporary AI agents can repeatedly discover, parse, summarize, and evaluate overlapping documents even when the resulting evidence is reusable. This pattern may waste network requests, CPU processing, model inference, and human verification; however, the act of web retrieval itself is not necessarily GPU-intensive. We propose *evidence-preserving research amortization* (EPRA), a domain-scoped architecture in which source-grounded research outputs are versioned as reusable knowledge packets and refreshed incrementally. We formalize the aggregate cost of independent research and packet-mediated research, derive a break-even condition dependent on fixed preparation cost, effective retrieval eligibility, fallback overhead, and maintenance, and introduce quality and freshness constraints that prevent cost reduction from becoming an excuse for unsupported answers. In a *fully synthetic* illustration, a 70% effective reuse rate produces break-even after 481 requests and a 38.6% aggregate cost reduction at 2,000 requests. These figures are algebraic consequences of illustrative assumptions, *not* measurements of Signal Atlas, GPU energy, or production AI systems. We distinguish the proposed research-workflow integration from established retrieval-augmented generation, semantic answer caching, key–value cache reuse, and model routing. Finally, we describe an experimental program comparing independent research, conventional retrieval augmentation, and packet-mediated retrieval on matched tasks, while measuring answer validity, citation fidelity, freshness, latency, and instrumented energy use. The contribution is a falsifiable systems thesis and reproducible evaluation design, not a claim of a new caching primitive or a proven energy saving.

Keywords: research amortization; retrieval-augmented generation; evidence provenance; agent systems; inference efficiency; incremental knowledge maintenance; scientific reproducibility.

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1 Introduction

Large language model (LLM) systems may spend substantial work acquiring and processing supporting information before answering a question. For agentic applications, a request can induce web queries, document downloading, extraction, chunking, embeddings, model-based reading, verification, and final synthesis. When similar questions recur across users, agents, or time, repeating the entire research chain is an avoidable candidate cost. Yet merely storing documents is not sufficient: source versions change, claims conflict, retrieved fragments omit caveats, and cached answers can become wrong.

Thesis. *For a bounded, recurring research workload, a shared, source-grounded and incrementally maintained knowledge layer can reduce total lifecycle computation relative to repeated independent research, provided its amortized build-and-maintenance cost plus per-query retrieval, verification, generation and fallback cost is smaller, subject to explicit quality and freshness requirements.*

The qualifier matters. It is incorrect to assume every AI request scrapes the web, that all web search uses GPUs, or that a local database is invariably cheaper. Search requests commonly consume network and CPU resources; accelerator use may arise primarily from repeated model-based reading, token processing, reranking, generation, and multi-step verification. Some local indexing and embedding pipelines also use accelerators. Accordingly, this paper treats *total computational cost* and *accelerator cost* as separate quantities rather than interchangeable labels.

We call the proposed integration *evidence-preserving research amortization* (EPRA), and the shared retrieval-and-governance service a *research amortization layer* (RAL). The research objects are *evidence packets*: compact but auditable source-linked representations of claims, assumptions, methods, contradictions and update status. The term is proposed for the end-to-end engineering practice; it is not claimed as the invention of RAG, semantic caching, knowledge graphs or incremental database maintenance.

1.1 Contributions and boundaries

This working paper offers: (i) a resource-accounted cost model; (ii) an analytic break-even threshold and reuse-prioritization rule; (iii) a freshness and verification formulation; (iv) a packet and fallback architecture motivated by the Signal Atlas domain; and (v) a preregistrable comparative experiment. It presents no new measured performance or verified physical law. Its immediate domain is public scientific and historical research relating to signals, electromagnetic science and published communications technology. It is not a proposal for intercepting protected communications or producing operational military targeting data.

2 Prior Art and the Research Gap

Retrieval-augmented generation (RAG) combines parametric model knowledge with a non-parametric corpus and was formalized as a general-purpose architecture by Lewis

and colleagues [1]. It establishes that grounding model outputs in external knowledge can improve knowledge-intensive tasks. It does not itself guarantee that evidence discovery and validation are shared across agents or that knowledge freshness and legal reuse are governed across research cycles.

Semantic answer caching, such as GPTCache [2], stores prior model responses for similar queries. EPRA differs in its primary unit of reuse: an evidence-backed packet, not merely a fluent answer. A packet may support multiple tasks and models, with a provenance trail that allows the final answer to be recomputed or challenged. Earlier work on *model routing* has also demonstrated large task-dependent differences in query cost; FrugalGPT shows that adaptive cascades can improve cost-quality tradeoffs in particular experimental settings [3]. Its reported savings are not claimed as EPRA results.

LLM inference caching represents a separate layer. PagedAttention and vLLM improve key–value (KV) memory management [4]; SGLang's RadixAttention reuses intermediate inference states [5]. CacheBlend reuses cached knowledge representations with selective recomputation [6], while RAGCache exploits recurring retrieved documents and pipelines search with decoding [7]. These methods help show that repeated contexts are computationally important, but KV-state reuse and evidence validation are different mechanisms. EPRA could complement, rather than replace, such serving optimizations.

Contextual retrieval can improve passage relevance [10], while the Lost in the Middle study warns that more context can impair the use of relevant evidence [8]. Inference efficiency also varies with GPU hardware, sequence length, batching, and serving configuration [9]. Green AI treats compute efficiency as an evaluation concern [12]. Accordingly, this proposal measures energy directly; fewer web searches alone do not establish lower GPU energy.

Table 1: Relationship to established methods. The proposed distinction is an engineering combination, not a claim of invention of component techniques.

Method	Usually reused	Primary missing guarantee for this proposal
Conventional RAG	Text passages or embeddings	Cumulative research review and quality-controlled packet admission.
Semantic caching	Finished response to a similar question	Reusable source-level evidence for new questions or changed assumptions.
KV / prefill caching	Inference intermediate states	Source provenance, scholarly interpretation, and rights verification.
Model routing	Selection of a lower-cost model/path	Shared research provenance and incremental refresh.
EPRA (proposed)	Dated evidence packets, claim graph, review status	Must still demonstrate net gains over strong RAG and cache baselines.

3 Formalization of Repeated Research

3.1 Workload, resources, and evidence packets

Let $\mathcal{Q} = \{q_1, \dots, q_N\}$ be a workload over horizon T, drawn from a possibly drifting distribution D_t. Each query may require a set of claims $\mathcal{C}(q)$ and supporting source material. An evidence packet K_i contains a version identifier, source URLs and document revisions, bounded claim statements, their supporting spans or locators, applicable units and assumptions, rights metadata, verification status, review timestamps, contradictory evidence, and a content hash. An answer is *eligible for local reuse* only when the packet is relevant, current enough, sufficient to answer the query, and passes the evidence-policy gate.

A cost is a vector, not inherently a single scalar:

$$\mathbf{c} = (c_{\text{net}}, c_{\text{cpu}}, c_{\text{gpu}}, c_{\text{store}}, c_{\text{audit}}), \qquad C = w^{\mathsf{T}}\mathbf{c}. \tag{1}$$

Here w supplies explicit exchange rates into dollars or another chosen unit. GPU-joules, wall time, requests, dollars and human labor hours should otherwise be reported as separate metrics. A hosted API token charge is a price proxy, not a direct physical energy measurement.

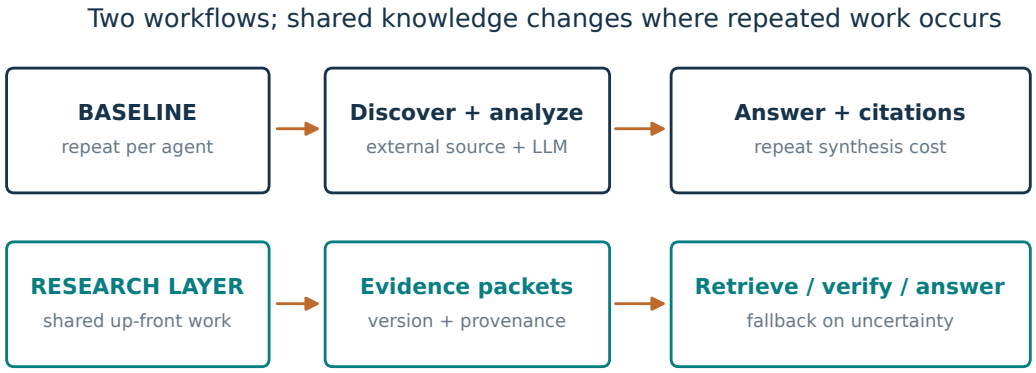


Figure 1: Architectural distinction between repeated independent research (upper path) and an evidence-preserving shared layer (lower path). Diagram is original to this paper.

3.2 Independent and amortized cost

For a simplified homogeneous workload, let $b > 0$ denote expected end-to-end cost per request when research is performed independently, including retrieval, extraction, necessary LLM inference, validation and response composition. Let $k(T)$ be the shared construction, indexing, review and refresh cost accrued over horizon T. Let h be per-query cost after an *eligible* packet hit, m the cost after an ineligible packet lookup and fallback, and $p \in [0, 1]$ the effective probability of eligible reuse. In general m can

exceed b because fallback includes an unsuccessful lookup.

$$C_{\text{base}}(N,T) = Nb, \tag{2}$$

$$C_{\text{RA}}(N,T) = k(T) + N\left[ph + (1-p)m\right]. \tag{3}$$

The effective hit rate is not merely vector-search recall. It incorporates whether the candidate packet is authorized, source-grounded, within the time scope, evidence-

Let G denote the joint event that a packet is in scope, authorized, retrieved, fresh, evidence-sufficient, and accepted by the answer-quality gate. Define

$$p = Pr(G) \tag{4}$$

Estimate this end-to-end probability directly. Any decomposition requires the full conditional chain rule; marginal rates cannot generally be multiplied.

Proposition 1 (Amortization break-even)**.** *For $N > 0$, define the expected per-query saving*

$$\Delta = b - \left[ph + (1-p)m\right]. \tag{5}$$

If $\Delta \leq 0$, this model yields no positive finite break-even workload. If $\Delta > 0$, shared packet research is cheaper precisely when

$$N > \frac{k(T)}{\Delta}. \tag{6}$$

Proof. Subtract (3) from (2); the net saving is $N\Delta - k(T)$. It is strictly positive if only if $\Delta > 0$ and $N > k(T)/\Delta$. □

When $m > h$, the corresponding requirement on reuse probability for a fixed N is

$$p > \frac{m - b + k(T)/N}{m - h}. \tag{7}$$

A high fraction of relevant questions is insufficient if packet quality is low, verification cost is large, or maintenance dominates.

3.3 Heterogeneous queries and reuse concentration

Actual queries are not identical. With topic or packet family i, expected future query volume n_i, net avoided independent work s_i, packet construction cost k_i and ongoing refresh/verification burden r_i, a first-order prioritization score is

$$V_i = n_i s_i - k_i - r_i. \tag{8}$$

This score is meaningful only after legal and evidence constraints pass. Dependencies, shared retrieval infrastructure and correlated query families may make item values non-additive. More generally, select admissible packets S to maximize $\sum_{i \in S} V_i$ under storage, labor, security and quality budgets; test interactions empirically. A broad but rarely queried corpus can be less efficient than a narrow corpus with high validated reuse.

4 Quality, Freshness, and the Limits of Compression

4.1 Quality is a hard constraint

Let Q_A and Q_B be predeclared task-quality measures for the amortized and strongest comparator systems. The scientific claim is a *constrained* optimization:

$$\begin{aligned} &\min_{\pi,\, S,\, \tau} \quad \mathbb{E}\left[C_{\mathrm{RA}}(N,T)\right] \\ &\text{subject to} \quad Q_A \geq Q_B - \varepsilon, \\ &\qquad\qquad F_A \geq F_{\min}, \quad R_{\text{citation}} \geq R_{\min}, \end{aligned} \tag{9}$$

where π is the query-routing policy, S the admitted packet set, τ a refresh schedule, F a freshness score, and R_{citation} the fraction of material claims supported by correct retrievable citations. The tolerance ε must be established before analysis; a fast but unsupported answer is not a successful optimization.

Packet *compression* here means concise, navigable representation, not wholesale deletion of the underlying evidence. A packet's summary should never replace a resolvable source identifier, exact publication version, conditions of applicability, or disconfirming material. In general there is no universal sufficient statistic for arbitrary future research questions. A compression scheme must be benchmarked against a declared question family, with its own omission/error rate.

4.2 A provisional refresh model

Suppose a source undergoes potentially material changes as a Poisson process with rate λ_i per unit time. If a packet was checked a time units ago, the probability that *at least one relevant source change* occurred is

$$P(\text{changed by } a) = 1 - e^{-\lambda_i a}. \tag{10}$$

This is a *model assumption*, not a claim that actual scientific sources change as Poisson processes. Under periodic checks spaced τ_i apart, small $\lambda_i \tau_i$ yields mean stale exposure approximately $\lambda_i \tau_i / 2$. If u_i is update cost and γ_i is the loss rate associated with stale exposure, the approximate cost per unit time is

$$J_i(\tau_i) \simeq \frac{u_i}{\tau_i} + \frac{\gamma_i \lambda_i \tau_i}{2}, \qquad \tau_i^* = \sqrt{\frac{2u_i}{\gamma_i \lambda_i}}. \tag{11}$$

Equation (11) is a heuristic optimum under the stated assumptions, and is not used for safety-critical, rapidly changing, or legally restricted claims without additional constraints. Event-driven invalidation, manual review and source retractions may supersede periodic refresh schedules.

4.3 Evidence admission and packet governance

A candidate should pass: canonical-source identification; provenance and licensing; claim-to-evidence linkage; applicable time and jurisdiction; independent corrobora-

tion when warranted; contradiction search; and review approval. An agent may propose candidates but must not silently promote them to verified knowledge. The W3C PROV-O standard supplies a formal vocabulary for entities, activities and agents in such a lineage [11]. A versioned store should preserve previous packets and make rollback possible. The same controls are relevant for educational and historical archives such as Signal Atlas.

5 Illustrative Numerical Model

This section supplies a completely reproducible *thought experiment*. All figures are synthetic *cost units*; they are not dollar costs, watt-hours, benchmark measurements, or measured Signal Atlas outcomes. We choose round numbers to make the break-even mechanism inspectable.

Table 2: Illustrative cost assumptions for a fixed observation horizon. GPU entries are a synthetic component of overall cost units, not measured energy.

Quantity	Value	Interpretation
b	180	Independent full research and answer per request.
h	45	Successful local packet retrieval, validity check, and answer.
m	190	Failed packet route plus conventional fallback.
$k(T)$	44,000	Initial build 40,000 plus horizon maintenance 4,000.
p	0.70	Effective eligible packet hit probability.
N	2,000	Number of evaluated requests.
GPU slice (b, h, m)	120, 25, 125	Separately declared illustrative accelerator component.
GPU fixed slice	10,000	Portion of shared setup/maintenance charged to GPU.

At $p = 0.7$, the variable per-request cost is $0.7(45) + 0.3(190) = 88.5$. Hence $\Delta = 91.5$ and the smallest integer number of requests yielding strictly positive net savings is $\lfloor 44{,}000/91.5 \rfloor + 1 = 481$. At $N = 2{,}000$:

$$C_{\text{base}} = 2{,}000(180) = 360{,}000, \tag{12}$$

$$C_{\text{RA}} = 44{,}000 + 2{,}000(88.5) = 221{,}000, \tag{13}$$

$$\text{modeled reduction} = 1 - \frac{221{,}000}{360{,}000} = 38.61\%. \tag{14}$$

Correction for strict thresholds: the reported integer threshold is $\lfloor k/\Delta \rfloor + 1$; at 50% reuse this equals 705, not a non-strict rounded value. The source code included with this paper computes thresholds from unrounded quantities.

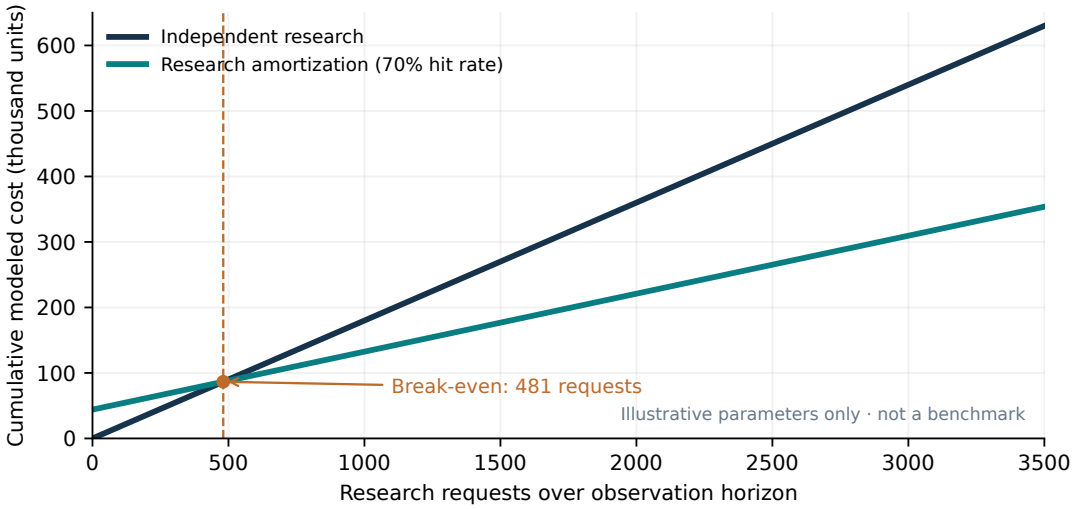


Figure 2: Cumulative cost under illustrative assumptions. Break-even is a mathematical consequence of the chosen cost model, not a measured throughput crossover.

Table 3: Sensitivity to effective packet reuse at $N = 2{,}000$. Figures are computed entirely from the illustrative model.

Eligible reuse	Cost / query	Break-even requests	Aggregate saving
30%	146.5	1,314	6.4%
50%	117.5	705	22.5%
70%	88.5	481	38.6%
90%	59.5	366	54.7%

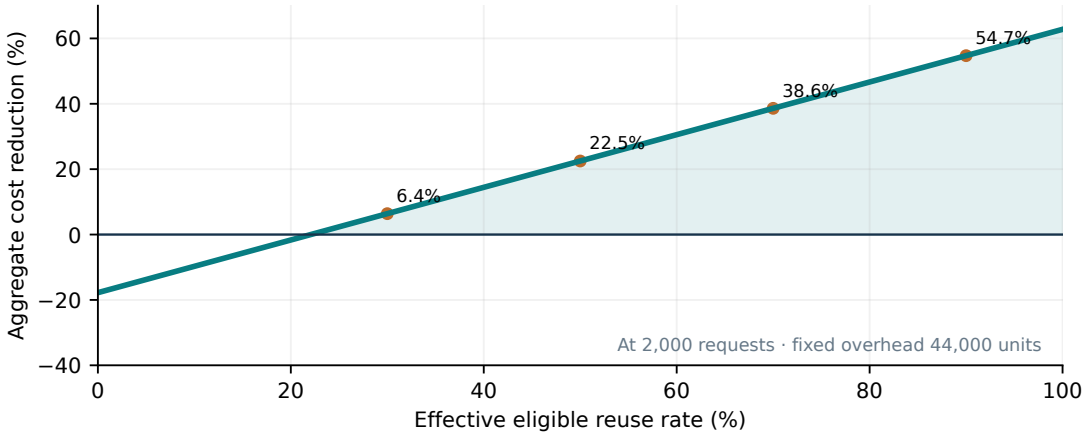


Figure 3: At fixed workload size, net savings depend on *effective* reuse rather than index size. Low reuse can make the architecture net-costly.

5.1 What the example does and does not show about GPUs

If the independent workload's modeled accelerator component is 120 per request, the packet hit is 25, the fallback is 125, and the fixed GPU share of construction is 10,000, then the modeled GPU subtotal at 2,000 requests is

$$G_{\mathrm{RA}} = 10{,}000 + 2{,}000\,[0.7(25) + 0.3(125)] = 120{,}000, \tag{15}$$

compared with $G_{\mathrm{base}} = 2{,}000(120) = 240{,}000$. This is a *50% arithmetic reduction in hypothetical GPU cost units*. It cannot be translated into a 50% real GPU electricity reduction without measuring GPU power over time, accounting for utilization, batching and idle energy. In many practical systems, extraction may be CPU-bound and a local vector search may incur its own GPU costs. *Inference energy is workload-dependent*, not a fixed linear function of token count [9].

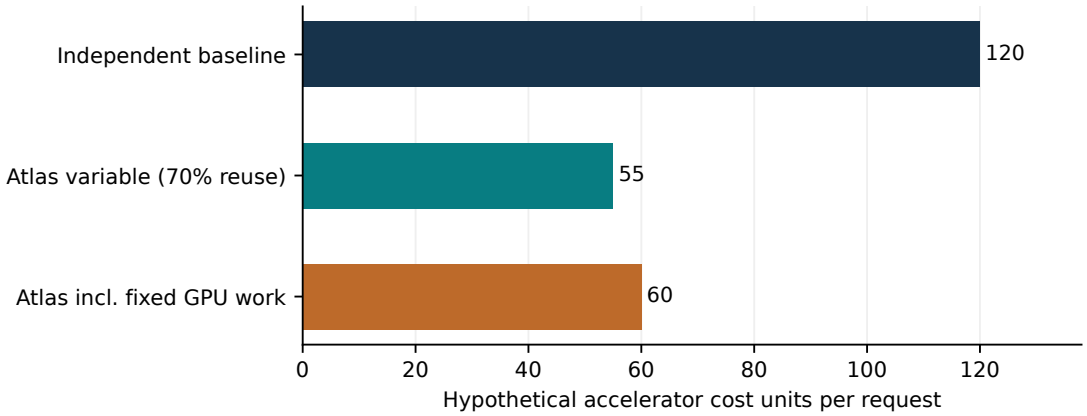


Figure 4: Accelerator-component accounting under declared synthetic assumptions. These bars are not joules, GPU-hours, telemetry or measured financial costs.

6 Implementation Architecture for Signal Atlas

Signal Atlas offers a bounded reference domain: electromagnetic science, public spectrum-management literature, technical signal processing, and documented historical communications intelligence. The following architecture is therefore a proposed *experimental testbed*, not a deployed compute-saving service.

1. **Source registry.** Preserve original URL/DOI, publisher, date, version, language, permitted reuse, source type and immutable content hashes for legitimately retained files. Distinguish a reference card from locally held full text.
2. **Evidence extraction.** Extract bounded propositions, equations and constraints; keep page, section or exact original locator, competing interpretations and dates of verification. Avoid turning a paper abstract into a confirmed experimental result.
3. **Admitted packets.** Store compact citation-rich research objects in a searchable index (lexical and/or vector) with provenance graph edges, source rights and quality status. Reject or quarantine speculative findings until reviewed.

4. **Query routing.** Determine scope, locate packets, test evidence sufficiency and freshness, assemble minimal explanatory context; when checks fail, fall back to live authorized research and propose a candidate update.
5. **Update lifecycle.** Recheck sources according to time sensitivity and material change signals. Maintain versioned deltas, an audit log, test fixtures and rollback instead of rebuilding the full corpus every run.

Each packet records identity and version, bounded claims, source references, equations and assumptions, review status, freshness dates, rights and a content hash (see Appendix B). A query can reuse only evidence it is entitled to see; multi-user systems need permission-scoped isolation. Neither a matching embedding nor a valid checksum proves that the factual content is correct.

6.1 A non-monotonicity warning

More stored evidence does not necessarily improve reasoning. Long passages can inflate prefill work and place relevant material in positions that models use poorly [8]. The packetizer should optimize evidence sufficiency and citation reliability, not raw compression ratio. A packet with fewer tokens but missing key exceptions can be *more expensive* after downstream verification errors or human corrections.

7 Testable Hypotheses and Experimental Protocol

7.1 Hypotheses

H1 / Total cost
: At sufficiently high effective reuse and fixed domain scope, EPRA reduces measured total per-query lifecycle cost versus independently repeated live research.

H2 / Accelerator use
: Conditional on matched models and comparable output quality, EPRA reduces aggregate GPU-joules from repeated document-reading and long-context inference; the effect is workload-dependent and may be zero.

H3 / Quality noninferiority
: Evidence packet routing achieves answer quality within a preregistered tolerance of the best feasible comparator, with no material deterioration in citation correctness or freshness.

H4 / Incremental maintenance
: When source changes are sparse, versioned source-delta processing uses fewer resources than periodic full-corpus re-ingestion at comparable freshness.

H5 / Shared reuse
: Cross-agent reuse is more valuable than single-agent reuse only when source permissions, trust boundaries, query overlap and verification costs allow it.

7.2 Comparison arms and task construction

A credible test must go beyond comparing EPRA to an intentionally inefficient system. Use, at minimum, four arms: **A** independent authorized web research with identical answer requirements; **B** conventional RAG over the *same available source corpus*; **C** RAG plus a strong available caching or context-optimization method; and **D** EPRA with evidence packets, review gates and controlled fallback. An ablation **E** removes packet verification or refresh to reveal their true cost and their effect on factual errors. Caching baselines should be configured fairly and reported transparently.

For an initial pilot, propose 600 queries partitioned into three balanced groups: stable reference questions, change-sensitive/publication-version questions, and novel questions with little or no packet coverage. Sample across subject families and question complexity. Reserve entire source families/topics for a held-out partition to avoid near-duplicate query leakage. Use the same model versions and comparable decoding configuration; when stochasticity remains, repeat with fixed and varied seeds. Scale beyond 600 if statistical power demands it.

7.3 Metrics and instrumentation

Table 4: Measurement plan. Cost and quality must be reported together.

Outcome	How measured
Primary quality	Blind human rubric and domain-expert review; factuality, completeness, correct units, and source support.
Citation fidelity	Claim-level supported-citation precision/recall, source-version accuracy and invalid-reference rate.
Compute cost	Model input/output tokens, embedding/reranker work, CPU-seconds, GPU execution time and full pipeline wall time.
Energy	Instrumented GPU energy (device counters or calibrated power traces), CPU energy when measurable, network/storage overhead if observed; report missing data.
Freshness	Source-change detection delay, fraction of responses relying on stale evidence and time to invalidate packets.
Workflow efficiency	Effective packet-hit rate, fallback frequency, p50/p95 latency, reuse count, build/maintenance cost, human review hours.

Physical energy should be measured as $E = \int P(t)\,dt$ with a defined system boundary, distinguishing device-level GPU energy from facility-level electricity (cooling and idle overhead). Hosted-token billing is not a substitute for this measurement. Report p50 and p95 latency separately from cumulative cost. Evaluate quality with a blinded adjudication protocol where possible; a model rating itself is insufficient.

7.4 Statistical analysis and stopping rules

Preregister the quality metric and a non-inferiority margin (e.g., an illustrative 2-percentage-point tolerance, subject to power analysis). Pair results by query and cluster bootstrap by topic or source family to avoid inflated significance from paraphrases. Report effect sizes, 95% confidence intervals, energy and latency distributions, failure modes, and negative results. Time-sensitive queries need chronological splits to

prevent test-time future evidence from leaking into static packets. Do not select only easy/high-hit tasks after seeing outcomes. If noninferiority fails, the system is *not demonstrated* to be an acceptable efficiency replacement even if it saves cost.

8 Threats to Validity and Failure Cases

Low reuse. One-off questions, highly diverse domains and novel discoveries may not amortize indexing costs. **Fast changing sources.** Markets, emergency updates, policy changes or newly released research can raise refresh work and stale-answer risk. **Quality collapse.** An overly compressed packet can omit a theorem's assumptions or a contrary experiment. **Hidden inference overhead.** Local embedding, reranking, permissions and trust verification may cost more than direct retrieval. **Energy rebound.** Reduced latency can encourage greater query volume, offsetting absolute energy savings. **Data rights and isolation.** Public readability does not imply permission to mirror documents, and reusable packets must not leak private material across principals.

One especially serious threat is *circular corroboration*: multiple agents may cite a shared synthesis that ultimately rests on one unverified document. The packet's provenance graph must retain independent source lineages and distinguish official self-description from independent empirical validation. Agent-generated candidate research must not be admitted automatically into the trusted store.

9 Research Agenda and Conclusion

EPRA's mathematical thesis is modest but consequential: reusable, evidence-preserving knowledge can be cost-effective when its fixed and recurring maintenance burden is dominated by cumulative avoided work, *and* when quality survives compression and routing. The benefit is not inherent to a database, nor is web scraping inherently a GPU burden. Real savings depend on workload overlap, long-context inference costs, local index overhead, source drift, verification expense and operational constraints.

The next research cycle should instrument one bounded Signal Atlas benchmark; compare it to fair RAG, cache and live-research baselines; verify the cost vectors rather than assuming them; and publish both failed and successful workloads. If the experiment validates the hypotheses, a follow-up paper can replace the present symbolic and synthetic results with replicated cost, accuracy, latency and energy measurements. Until then, *research amortization is a falsifiable systems-design proposal backed by related prior work, not a demonstrated scientific breakthrough.*

A Derivation and Reproducibility Notes

The exact net saving in the homogeneous model is $S(N) = Nb - [k(T) + N(ph + (1-p)m)]$. For $m > h$ and a fixed horizon, $\partial S/\partial p = N(m-h) > 0$, explaining why improving *eligible* packet coverage can be valuable. But if improving p increases $k(T)$ nonlinearly (e.g., expensive human verification), the true optimum may be interior.

For the demonstration, b = 180, h = 45, m = 190, k = 44,000, N = 2,000; the separate synthetic GPU slice is (120, 25, 125), with fixed GPU overhead 10,000. These invented parameters illustrate the proof and must not be cited as field observations. The companion model.py in this release bundle generates the plotted curves and sensitivity CSV from these constants.

A separate quality and cost noninferiority decision can be formalized by requiring $\mathrm{LCB}_{95\%}(Q_A - Q_B) > -\varepsilon$ and $\mathrm{UCB}_{95\%}(C_A - C_B) < 0$ after the stated statistical corrections. These criteria should be registered in advance and interpreted jointly. They are more defensible than reporting a single cost-saving percentage without error bounds.

B Minimal Evidence Packet Schema (Illustrative)

The following schematic fields are not a deployed standard, and all example identifiers are placeholders:

```
{
 "packet_id": "atlas:science:example:0001",
 "version": "0.1.0",
 "domain": "electromagnetic-signal-science",
 "claim_ids": ["claim:verified-in-source:0001"],
 "source_refs": [{"url": "https://example.org/paper",
                  "version": "v1", "section": "2"}],
 "assumptions": ["bounded domain", "source freshness"],
 "contradictions": [],
 "rights": "metadata-and-link-only",
 "review_status": "candidate",
 "checked_at": "2026-11-02",
 "expires_at": null,
 "content_hash": "REPLACE_WITH_ACTUAL_HASH"
}
```

An evidence packet should be rejected or routed to external research if its sources are missing, its permission scope is violated, its applicable assumptions fail, its freshness policy is unmet, or its content is insufficient for the question. An actual published schema would require versioned validation rules, threat modeling, and test fixtures.

C Terms and Units

N, T	Number of queries and the observation horizon.
b, h, m	Per-query independent, eligible packet-hit, and fallback costs, respectively, in a defined common unit.
$k(T)$	Shared build, ingestion, verification and maintenance cost over T.
p	Effective probability of evidence-sufficient, authorized, fresh packet reuse.
Δ	Expected variable cost avoided per query, excluding fixed overhead.
Q, F, R	Answer quality, freshness and supported-citation fidelity metrics.

λ_i, τ_i Simplified source-change hazard and interval between checks.

GPU energy
Measured watt-seconds (joules), not merely a count of GPUs, a paid API fee, or synthetic cost units.

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APPENDIX D

Related Work and Contribution Boundary

Long-term agent memory is a direct neighboring area. Generative Agents record experience, retrieve salient memories, and use reflection to support future behavior [13]. MemoryBank updates and retrieves conversational memories for long-running interactions [14]. MemGPT manages virtual context across memory tiers, including document analysis and multi-session dialogue [15]. These systems establish that persistent memory, selective retrieval, and memory management predate the present proposal.

The candidate distinction in evidence-preserving research amortization (EPRA) is narrower: a source-evidence packet and its admission lifecycle. The packet records claim-to-source spans, source version and time or jurisdiction scope, rights, contradictions, verification status, freshness policy, and audit history; failed gates trigger refresh, review, or authorized external research. Its intended reuse unit supports re-examination across tasks rather than only recalling prior user interactions or caching a finished answer. EPRA can be implemented with established retrieval and memory components.

This is a systems-integration and evaluation hypothesis, not a priority claim. No exhaustive prior-art search has established that every component is new. A 2026 survey is included as a preprint orientation and should be checked against the version current at submission [16]. Any empirical contribution depends on a controlled comparison against retrieval, caching, and agent-memory baselines with quality, provenance, freshness, and resource outcomes reported together.

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APPENDIX E

Preregistration and Reproducibility Protocol

E.1 Scope and estimand

Before confirmatory evaluation, freeze the domain, source corpus, task-generation rules, inclusion criteria, software versions, model identifiers, prompts, index and retriever settings, hardware, and system boundary. The unit of analysis is a task query. Group queries into three declared strata: stable-reference questions, version-sensitive or change-sensitive questions, and novel or low-coverage questions. Hold out complete topic or source families to limit leakage.

The primary cost outcome is a preregistered vector, not an unweighted scalar: GPU energy in joules, CPU time, wall-clock latency, network transfer, storage, hosted API charges, and human review time. Include retrieval, indexing, refresh, verification, fallback, answer generation, and maintenance over the stated horizon. If a single aggregate is required, publish its conversion weights and rationale before examining outcomes. Report each component even when an aggregate is used.

E.2 Comparison arms and sampling

Compare four systems on identical queries and the same authorized source universe: (A) independent live research; (B) conventional retrieval-augmented generation; (C) retrieval with a strong available caching or context-optimization baseline; and (D) EPRA with versioned packets, verification, freshness checks, and fallback. Add an EPRA ablation that removes the preregistered packet gate or gates, so any cost change can be weighed against the errors those gates prevent. Keep output budgets and task information comparable, and document unavoidable differences.

The 600-query figure in the main text is a pilot target, not a claim of adequate confirmatory power. Estimate the confirmatory sample size from a separate pilot using the prespecified paired non-inferiority margin, variance, type-I error, and target power; freeze the calculation before the confirmatory run. Pair all arms by query, randomize run order, and reserve entire source families or topics for the held-out split. Do not tune on that split.

E.3 Instrumentation boundary

Measure accelerator energy with device telemetry or calibrated power traces over a declared interval. State whether the boundary includes host CPU, memory, cooling, idle draw, and shared-service overhead; report missing or estimated components separately. GPU utilization, token counts, and API prices are not substitutes for measured energy. Record device model, driver, power cap, batch size, sequence length, concurrency, warm-up, and idle conditions.

APPENDIX E (CONTINUED)

Analysis, Decision Rules, and Release

E.4 Quality and provenance outcomes

Predefine one primary answer-quality rubric and an explicit non-inferiority margin before confirmatory evaluation. Have assessors who are blind to system arm score factual support, completeness, correct units, and answer relevance using a written adjudication guide. Score claim-level citation precision and recall, source/version accuracy, freshness violations, unsupported claims, and rights or scope failures as separate outcomes. Resolve disagreements by a preregistered procedure and report assessor agreement.

E.5 Statistical analysis

Use paired differences at query level and account for topic or source-family clustering. Report effect sizes and 95% confidence intervals for the primary quality and cost outcomes; use a preregistered clustered bootstrap or a justified hierarchical model. Identify one primary comparison and outcome in advance, adjust or clearly label secondary analyses, and report failures and missing measurements. Do not use optional stopping or choose the reported task subset after inspecting results. A system that fails the quality non-inferiority rule cannot be declared an efficiency success solely because its measured cost is lower.

E.6 Logging and reproducibility

Publish a timestamped protocol, immutable task and source manifests, hashes, software and model versions, random seeds, configuration files, analysis code, instrumentation calibration notes, and per-query results where permissions allow. Log packet admission, evidence spans, freshness decisions, fallback events, verification failures, updates, and human interventions. Document licensing, privacy, retention, and any source restrictions; do not redistribute material without permission. If data cannot be released, provide hashes and a lawful replication route.

E.7 Interpretation and release status

Separate measured outcomes from model-generated illustrations. The numerical examples in this paper remain synthetic until replaced by a preregistered, instrumented study. Release negative findings and deviations alongside positive results. Report changes to the protocol with dates and reasons, and keep the analysis independent of the final test set. This appendix is a prospective protocol proposal; it is not evidence that the experiment has been run.